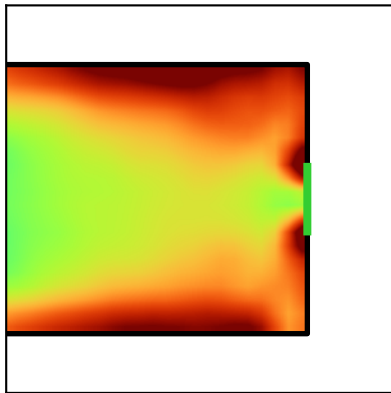
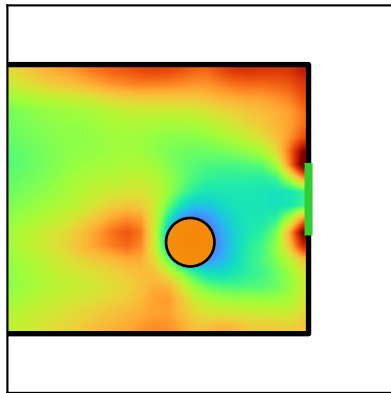


Steady-state crowd contact force at exit, $\theta = 90^\circ$

(a) no obstacle



(b) single pillar, offset (2.2 m)



(c) two-pillar gate (1.1 m gap)

